

Vertiv Liebert ItON 600VA CX

Very few people might have heard of Vertiv when they're out to purchase a UPS (Uninterrupted Power Supply). What might come as a surprise is that Vertiv is actually a massive company that handles infrastructure and services for data centres. Liebert Corporation is a subsidiary which has been manufacturing power infrastructure systems for decades and the Vertiv Liebert ItON range of uninterrupted power supplies brings all their decades of experience into the consumer segment.

The Vertiv Liebert ItON 600VA CX is a line-interactive UPS aimed for home and small office applications. It's a pre-assembled unit which means that you can use it right out of the box without any issues. Often, there are UPS units which come with the battery disconnected which need to be plugged in before use. With improved battery technologies, there's no need to ship UPS units without the battery plugged in and that's the case with the Vertiv Liebert ItON 600VA CX as well.

As for the specifications, the input voltage range is between 140-300 Volts AC and the AC voltage regulation has a tolerance of about 10 per cent. If the UPS switches to battery mode, then the AC voltage range tolerance becomes 1 per cent which is pretty good for a line interactive UPS. The transfer time between AC to battery or vice-versa is typically between 2 to 6 milliseconds.

It's also a simulated sine wave UPS. For certain power supplies installed in desktop computers, this is a problematic combination. When you have a desktop PC built using an SMPS that has an Active PFC, then it's always advised to use a Pure Sine Wave UPS rather than a Simulated Sine Wave UPS. Using a Simulated Sine Wave UPS might cause the PC to reboot in such cases. However, Pure Sine Wave SMPS are usually more expensive and therefore rarer to come across. Nevertheless, you should



always check your SMPS before finding the right UPS to pair with it. Vertiv has line-interactive UPS units with Pure Sine Wave output as well, such as the Vertiv Liebert PSI5. Coming to the protection circuits, overload protection, discharge protection and overcharge protection. All in all, it's got a nice feature set considering the price bracket.

The Vertiv Liebert ItON 600VA CX has a rigid plastic body like most UPS in the price range. The front of the unit has a push-to-on button for switching the unit on and a little LED indicator for the status of the UPS right next to the power button. There are three output sockets on the back side of the UPS and each socket is the typical Type D 5 Ampere socket that's most popular in India.

There are ventilation slits on the top surface and bottom surface of the body. And the bottom has four plastic feet. We feel that they could have added rubber inserts in the feet considering that it has a pretty beefy transformer inside it.

Speaking of the insides. There is a Class B transformer meaning that it is rated for 130 degrees Celsius. So should there be heavy load on the UPS, then the transformer should be capable of handling a lot of heat that's typically generated under high-load

conditions. The main switching circuit is attached to the top side of the UPS and seems to be a fairly simple design manufactured by TAIWAN WAN NIEN ELECTRIC APPLIANCE CO., LTD. There are four YL303H-S-12VDC-1Z PCB relays with 10 Ampere switching capability for when the UPS needs to switch from AC power to battery backup. It follows a Buck and Boost

design to perform Automatic Voltage Regulation in the UPS to perform adjustments to the incoming AC voltage to stabilise the power that is being sent out.

Now if you know the power consumption of all the devices that you are plugging into the UPS, then you can calculate how long it will last with a simple calculation. If your connected devices are pulling about 120 Watts then the Vertiv Liebert ItON 600VA CX can provide up to 15 minutes of back up time. This is provided there is a continuous load of 120 Watts. High-end gaming PCs will draw a lot more power and will give you less back up time. But with UPS units, the primary objective is to give you enough time to save all your work and then safely shut down your PC.

We connected the UPS to a decently powerful gaming PC and fired it up. With the machine running a couple of applications, we switched off the main power to see how long the UPS could provide back up power. With nominal load, the unit provided about 19 minutes of back up which is quite decent. It gives plenty of headroom to actually complete whatever tasks that you may have running on the PC.

—Mithun Mohandas